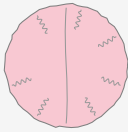


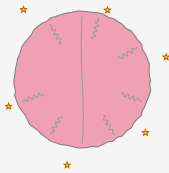
# Detecting AI-generated images: a journey

(ADAIS / wave\_1)



“Looks real to me.”

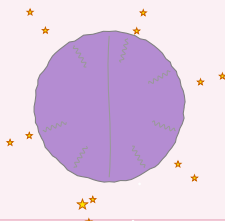
*(accuracy:  $\sim 50\%$ , confidently wrong)*



“I’ll fine-tune a ResNet-50 on the training set.”

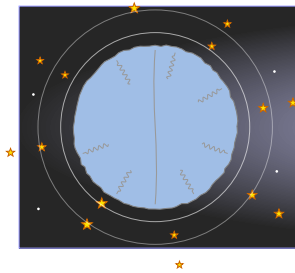
*(works on the training distribution.*

*fails the moment a new generator drops)*



“The radial FFT of the wavelet residual reveals up-sampling artefacts the generator can’t hide.”

*(now we’re talking. coarse acc: 0.72)*



“Fit the Poisson–Gaussian noise-level function, locate Bayer demosaicing peaks at the half-Nyquist, and reconstruct sub-pixel lateral chromatic aberration from a  $256 \times 256$  thumbnail.”

*(you cannot fake a lens you do not have)*

*The classifier doesn’t ask “does this look fake?”  
It asks “does this look like it came out of a camera?”*